

# $K$ -theoretic Catalan functions

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- Schubert calculus
- Catalan functions: a new approach to old problems
- $K$ -theoretic Catalan functions

# Overview of Schubert Calculus Combinatorics

## Geometric problem

Find  $c_{\lambda\mu}^\nu = \#$  of points in intersection of subvarieties in a variety  $X$ .

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Special basis of polynomials  $\{f_\lambda\}$  such that  $f_\lambda \cdot f_\mu = \sum_\nu c_{\lambda\mu}^\nu f_\nu$

# Overview of Schubert Calculus Combinatorics (cont.)

Combinatorial study of  $\{f_\lambda\}$  enlightens the geometry (and cohomology).

## Goal

Identify  $\{f_\lambda\}$  in explicit (simple) terms amenable to calculation and proofs.

## Geometric problem

Find  $c_{\lambda\mu}^\nu = \#$  of points in intersection of Schubert varieties  $\{X_\lambda\}_{\lambda \subseteq (n^m)}$  in variety  $X = \text{Gr}(m, n)$ .

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## Representatives

Special basis of Schur polynomials  $\{s_\lambda\}$  such that  $s_\lambda \cdot s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$  for Littlewood-Richardson coefficients  $c_{\lambda\mu}^\nu$ .

# Schur functions $s_\lambda$

## Example

*Semistandard tableaux*: columns increasing and rows non-decreasing.

|   |   |   |   |
|---|---|---|---|
| 5 |   |   |   |
| 3 | 4 |   |   |
| 2 | 3 |   |   |
| 1 | 2 | 2 | 5 |

|   |   |   |   |
|---|---|---|---|
| 8 |   |   |   |
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standard = no repeated letters

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Schur function  $s_\lambda$  is a “weight generating function” of semistandard tableaux:

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 2 | 3 | 3 | 2 | 3 | 3 | 2 | 3 |
| 1 | 1 | 1 | 1 | 1 | 2 | 3 | 1 |
| 1 | 1 | 2 | 2 | 3 | 3 | 3 | 2 |

$$s_{\square\square}(x_1, x_2, x_3) = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_2^2 + x_1 x_3^2 + x_2 x_3^2 + 2x_1 x_2 x_3$$

# Schur functions $s_\lambda$ (cont.)

## Pieri rule

Determines multiplicative structure:

$$s_r s_\lambda = \sum (1 \text{ or } 0) s_\nu$$

$$s_{\square} s_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} = s_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + s_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + s_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}}$$

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Since  $s_{\mu_1} \cdots s_{\mu_r} = s_{(\mu_1, \dots, \mu_r)} + \text{lower order terms}$ , subtract to get

$$s_{(\mu_1, \dots, \mu_r)} s_\lambda = \sum c_{\lambda\mu}^\nu s_\nu$$

for well-understood *Littlewood-Richardson coefficients*  $c_{\lambda\mu}^\nu$ .

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$$\mathfrak{S}_{s_i} = x_1 + \cdots + x_i$$

### Open Problem

Structure constants  $\mathfrak{S}_w \mathfrak{S}_u = \sum_v c_{wu}^v \mathfrak{S}_v$  have no tableaux description.

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| (Co)homology of Grassmannian           | Schur functions              |
| (Co)homology of flag variety           | Schubert polynomials         |
| Quantum cohomology of flag variety     | Quantum Schuberts            |
| (Co)homology of Types BCD Grassmannian | Schur- $P$ and $Q$ functions |
| (Co)homology of affine Grassmannian    | (dual) $k$ -Schur functions  |
| $K$ -theory of Grassmannian            | Grothendieck polynomials     |
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And many more!

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$$\begin{aligned}\Phi: QH^*(Fl_{k+1}) &\rightarrow H_*(Gr_{SL_{k+1}})_{loc} \\ \mathfrak{S}_w^Q &\mapsto \frac{s_\lambda^{(k)}}{\prod_{i \in Des(w)} \tau_i}\end{aligned}$$

where  $s_\lambda^{(k)}$  is a  $k$ -Schur symmetric function and  $Gr_{SL_{k+1}}$  is the “affine Grassmannian.”

Upshot

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## Upshot

Computations for (quantum) Schubert polynomials can be moved into symmetric functions.

# $k$ -Schur functions

- $s_{\lambda}^{(k)}$  for  $\lambda_1 \leq k$  a basis for  $\mathbb{Z}[s_1, s_2, \dots, s_k]$  (Lapointe et al., 2003).

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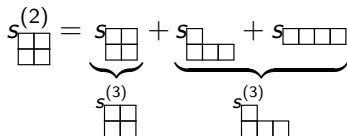
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$$s_{\lambda}^{(2)} = s_{\lambda} + s_{\lambda} + s_{\lambda}$$

The diagram illustrates the branching of the Schur function  $s_{\lambda}^{(2)}$  into three Schur functions  $s_{\lambda}^{(3)}$ . The left side shows  $s_{\lambda}^{(2)}$  as a 2x2 square. The right side shows the sum of three 3x3 squares, each representing a different  $s_{\lambda}^{(3)}$ . The first square is the same as the 2x2 square. The second square has an additional box in the second row, first column. The third square has an additional box in the second row, second column. Brackets below the squares indicate they are all  $s_{\lambda}^{(3)}$ .

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The diagram shows the branching of the 2-Schur function  $s_{(2)}^{(2)}$  into three 3-Schur functions. On the left is a 2x2 square representing  $s_{(2)}^{(2)}$ . On the right is the sum of three Young diagrams: a 2x2 square ( $s_{(2)}^{(3)}$ ), a 2x1 and 1x1 squares ( $s_{(1,1)}^{(3)}$ ), and three 1x1 squares ( $s_{(1,1,1)}^{(3)}$ ). Brackets below the right side group the three terms under  $s_{(2)}^{(3)}$  and  $s_{(1,1)}^{(3)}$ .

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- (Lam et al., 2010) gives geometric interpretation,
- but no combinatorial interpretation of branching coefficients.
- Branching with  $t$  important for Macdonald polynomial positivity.
- Many conjecturally equivalent definitions.

- Schubert calculus
- **Catalan functions: a new approach to old problems**
- $K$ -theoretic Catalan functions

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Key: Catalan functions = large class of symmetric functions.

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$$s_{22} = (1 - R_{12}) h_{22} = h_{22} - h_{31}$$

$$s_{211} = (1 - R_{12})(1 - R_{23})(1 - R_{13}) h_{211}$$

$$= h_{211} - h_{301} - h_{220} - \text{red } h_{310} + \text{red } h_{310} + \underbrace{h_{32-1}}_{=0} + h_{400} - \underbrace{h_{41-1}}_{=0}$$

some terms cancel

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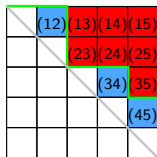
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$$s_{1^r}^\perp s_\lambda = \sum_{S \subseteq [1, \ell], |S|=r} s_{\lambda - \epsilon_S}$$

$$s_{1^2}^\perp s_{333} = s_{322} + s_{232} + s_{223}$$

# Root Ideals

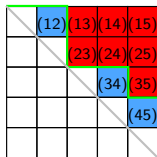
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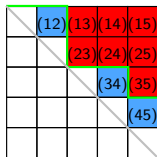
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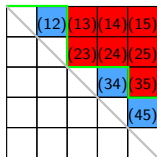
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## Intuition

Catalan functions interpolate between  $h_\lambda$  and  $s_\lambda$ .

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## Theorem (Blasiak et al., 2020)

For  $\Psi$  any root ideal and  $\lambda$  a partition,  $H(\Psi; \lambda)$  is Schur positive!

## $k$ -Schur root ideal for $\lambda$

$$\begin{aligned}\psi = \Delta^k(\lambda) &= \{(i, j) : j > k - \lambda_i\} \\ &= \text{root ideal with } k - \lambda_i \text{ non-roots in row } i\end{aligned}$$

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$$\Delta^4(3, 3, 2, 2, 1, 1) =$$

|   |   |   |   |   |   |  |
|---|---|---|---|---|---|--|
| 3 |   |   |   |   |   |  |
|   | 3 |   |   |   |   |  |
|   |   | 2 |   |   |   |  |
|   |   |   | 2 |   |   |  |
|   |   |   |   | 1 |   |  |
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$k$ -Schur is a Catalan function (Blasiak et al., 2019).

For partition  $\lambda$  with  $\lambda_1 \leq k$ ,

$$s_{\lambda}^{(k)} = H(\Delta^k(\lambda); \lambda).$$

# Key ingredient of branching proof

Dual vertical Pieri rule:  $s_{1^r}^\perp s_\lambda^{(k)} = \sum_\mu a_{\lambda\mu} s_\mu^{(k)}$  for  $\langle s_{1^r}^\perp f, g \rangle = \langle f, s_{1^r} g \rangle$ .

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|---|---|---|---|---|---|
| 3 |   |   |   |   |   |
|   | 3 |   |   |   |   |
|   |   | 2 |   |   |   |
|   |   |   | 2 |   |   |
|   |   |   |   | 1 |   |
|   |   |   |   |   | 1 |

$$\Delta^5(4, 4, 3, 3, 2, 2) =$$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 4 |   |   |   |   |   |
|   | 4 |   |   |   |   |
|   |   | 3 |   |   |   |
|   |   |   | 3 |   |   |
|   |   |   |   | 2 |   |
|   |   |   |   |   | 2 |

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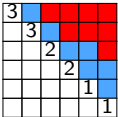
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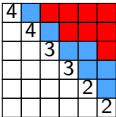
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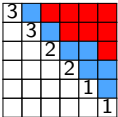
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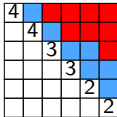
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Branching is a special case of Pieri:

$$s_\lambda^{(k)} = s_{1^\ell}^\perp s_{\lambda+1^\ell}^{(k+1)} = \sum_\mu a_{\lambda+1^\ell, \mu} s_\mu^{(k+1)}$$

- Schubert calculus
- Catalan functions: a new approach to old problems
- **$K$ -theoretic Catalan functions**

# Dual Grothendieck polynomials

- Inhomogeneous basis:  $g_\lambda = s_\lambda + \text{lower degree terms}$ .

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- Dual to Grothendieck polynomials  $G_\lambda$ : Schubert representatives for  $K^*(Gr(m, n))$

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2-bounded partitions  $\leftrightarrow$  3-cores

The diagram illustrates the Pieri rule for  $K$ - $k$ -Schur functions. It shows the product of a 1-strip ( $g_1$ ) and a 2-bounded partition ( $g_{211}^{(2)}$ ) resulting in the difference of two 2-bounded partitions ( $g_{2111}^{(2)} - 2g_{211}^{(2)}$ ). The partitions are represented as 5x5 grids of dots (red, blue, black) with some cells shaded gray to indicate the addition or subtraction of strips.

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No direct formula for  $g_{\lambda}^{(k)}$

## Solution

Find a formula for  $g_{\lambda}^{(k)}$  analogous to raising operator formula for  $s_{\lambda}^{(k)}$ .



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Requires an inhomogeneous refinement of Catalan functions.

# An Extra Ingredient: Lowering Operators

Lowering Operators  $L_j(f_\lambda) = f_{\lambda - \epsilon_j}$

$$L_3 \left( \begin{array}{|c|c|c|c|} \hline \text{red} & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array} \right) = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}, \quad L_1 \left( \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & \text{red} & \\ \hline \end{array} \right) = \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline \end{array}$$

## $K$ -theoretic Catalan function

Let  $\Psi, \mathcal{L} \subseteq \Delta_\ell^+$  be order ideals of positive roots and  $\gamma \in \mathbb{Z}^\ell$ , then

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# Affine $K$ -Theory Representatives with Raising Operators

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## Example

non-roots of  $\Psi$ , roots of  $\mathcal{L}$

|  |      |      |      |      |
|--|------|------|------|------|
|  | (12) | (13) | (14) | (15) |
|  |      | (23) | (24) | (25) |
|  |      |      | (34) | (35) |
|  |      |      |      | (45) |
|  |      |      |      |      |

$$\begin{aligned} K(\Psi; \mathcal{L}; 54332) \\ = (1 - L_4)^2 (1 - L_5)^2 (1 - R_{12}) (1 - R_{34}) (1 - R_{45}) k_{54332} \end{aligned}$$

Answer (Blasiak-Morse-S., 2020)

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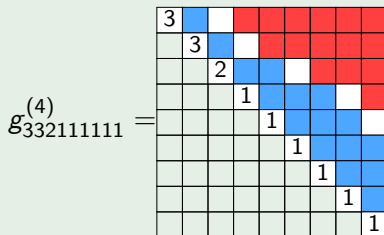
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Example



$$\Delta_9^+ / \Delta^4(332111111), \Delta^5(332111111)$$

# Pieri Rule Illustrated (Recurrences)

A “graphical calculus.”

$$g_1 g_{211}^{(2)}$$



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$$g_1 g_{211}^{(2)}$$

$$=$$

|   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|--|
| 2 |   |   |   |   |   |   |  |
|   | 1 |   |   |   |   |   |  |
|   |   | 1 |   |   |   |   |  |
|   |   |   | 0 |   |   |   |  |
|   |   |   |   | 0 |   |   |  |
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|   |   |   |   | 0 |   |   |
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$$=$$

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| 2 |   |   |   |   |   |   |
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|   |   | 1 |   |   |   |   |
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$$+$$

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|   | 1 |   |   |   |   |   |
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|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
|   |   |   |   | 0 |   |   |
|   |   |   |   |   | 0 |   |
|   |   |   |   |   |   | 1 |

$$+$$

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|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
|   |   |   |   | 0 |   |   |
|   |   |   |   |   | 0 |   |
|   |   |   |   |   |   | 1 |

$$=$$

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|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
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|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
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|   |   |   | 0 |   |   |   |
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|   |   |   |   |   | 0 |   |
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$$+$$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
|   |   |   |   | 0 |   |   |
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|   |   |   |   |   |   | 1 |

# Pieri Rule Illustrated (Straightening)

$$g_1 g_{211}^{(2)} =$$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
|   |   |   |   | 0 |   |   |
|   |   |   |   |   | 0 |   |
|   |   |   |   |   |   | 1 |

 $+$ 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
|   |   |   |   | 0 |   |   |
|   |   |   |   |   | 0 |   |
|   |   |   |   |   |   | 1 |

 $+$ 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
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|   |   |   | 0 |   |   |   |
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$$+$$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
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|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
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|   |   |   | 1 |   |   |   |
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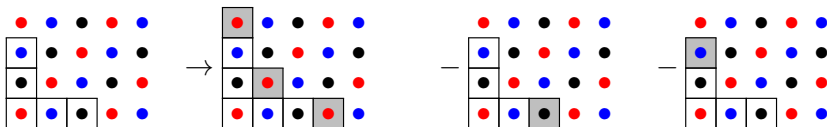
# Pieri Rule Illustrated (Straightening)

$$\begin{aligned}
 g_1 g_{211}^{(2)} &= \begin{array}{|c|c|c|c|c|c|c|} \hline 2 & & & & & & \\ \hline & 1 & & & & & \\ \hline & & 1 & & & & \\ \hline & & & 0 & & & \\ \hline & & & & 0 & & \\ \hline & & & & & 0 & \\ \hline & & & & & & 1 \\ \hline \end{array} + \begin{array}{|c|c|c|c|c|c|c|} \hline 2 & & & & & & \\ \hline & 1 & & & & & \\ \hline & & 1 & & & & \\ \hline & & & 0 & & & \\ \hline & & & & 0 & & \\ \hline & & & & & 0 & \\ \hline & & & & & & 1 \\ \hline \end{array} + \begin{array}{|c|c|c|c|c|c|c|} \hline 2 & & & & & & \\ \hline & 1 & & & & & \\ \hline & & 1 & & & & \\ \hline & & & 1 & & & \\ \hline & & & & 0 & & \\ \hline & & & & & 0 & \\ \hline & & & & & & 1 \\ \hline \end{array} \\
 &= \begin{array}{|c|c|c|c|c|} \hline 2 & & & & \\ \hline & 1 & & & \\ \hline & & 1 & & \\ \hline & & & 1 & \\ \hline & & & & 1 \\ \hline \end{array} - \begin{array}{|c|c|c|c|} \hline 2 & & & \\ \hline & 1 & & \\ \hline & & 1 & \\ \hline & & & 1 \\ \hline \end{array} - \begin{array}{|c|c|c|c|} \hline 2 & & & \\ \hline & 1 & & \\ \hline & & 1 & \\ \hline & & & 1 \\ \hline \end{array} \\
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Theorem (S. (thesis), 2021)

*For  $1 \leq d \leq k$ , set  $R_d = ((k + 1 - d)^d)$  to be the  $k$ -rectangle partition.*

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- $k$ -Rectangle Property fails for  $g_{\lambda}^{(k)}$ .



# Positivity of Catalan functions

Recall (Blasiak et al., 2020)

For  $\Psi$  any root ideal and  $\lambda$  a partition,  $H(\Psi; \lambda)$  is Schur positive.

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## Thank you!

- Anderson, David, Linda Chen, and Hsian-Hua Tseng. 2017. *On the quantum  $K$ -ring of the flag manifold*, preprint. arXiv: 1711.08414.
- Blasiak, Jonah, Jennifer Morse, Anna Pun, and Daniel Summers. 2019. *Catalan Functions and  $k$ -Schur Positivity*, J. Amer. Math. Soc. **32**, no. 4, 921–963.
- Blasiak, Jonah, Jennifer Morse, and Anna Pun. 2020. *Demazure crystals and the Schur positivity of Catalan functions*, preprint. arXiv: 2007.04952.
- Blasiak, Jonah, Jennifer Morse, and George H. Seelinger. 2020.  *$K$ -theoretic Catalan functions*, preprint. arXiv: 2010.01759.
- Chen, Li-Chung. 2010. *Skew-linked partitions and a representation theoretic model for  $k$ -Schur functions*, Ph.D. thesis.
- Fomin, Sergey, Sergei Gelfand, and Alexander Postnikov. 1997. *Quantum Schubert polynomials*, J. Amer. Math. Soc. **10**, no. 3, 565–596, DOI 10.1090/S0894-0347-97-00237-3. MR1431829
- Ikeda, Takeshi, Shinsuke Iwao, and Toshiaki Maeno. 2018. *Peterson Isomorphism in  $K$ -theory and Relativistic Toda Lattice*, preprint. arXiv: 1703.08664.
- Lam, Thomas. 2008. *Schubert polynomials for the affine Grassmannian*, J. Amer. Math. Soc. **21**, no. 1, 259–281.
- Lam, Thomas, Luc Lapointe, Jennifer Morse, and Mark Shimozono. 2010. *Affine insertion and Pieri rules for the affine Grassmannian*, Mem. Amer. Math. Soc. **208**, no. 977.
- Lam, Thomas, Anne Schilling, and Mark Shimozono. 2010.  *$K$ -theory Schubert calculus of the affine Grassmannian*, Compositio Math. **146**, 811–852.
- Lapointe, Luc, Alain Lascoux, and Jennifer Morse. 2003. *Tableau atoms and a new Macdonald positivity conjecture*, Duke Mathematical Journal **116**, no. 1, 103–146.
- Morse, Jennifer. 2011. *Combinatorics of the  $K$ -theory of affine Grassmannians*, Advances in Mathematics.
- Panyushev, Dmitri I. 2010. *Generalised Kostka-Foulkes polynomials and cohomology of line bundles on homogeneous vector bundles*, Selecta Math. (N.S.) **16**, no. 2, 315–342.



$$k_m^{(r)} = \sum_{i=0}^m \binom{r+i-1}{i} h_{m-i} = s_m(X+r),$$

a specialization of “multiSchur functions.” See, e.g., Lascoux-Naruse (2014).

$$k_{\gamma} = k_{\gamma_1}^{(0)} k_{\gamma_2}^{(1)} \dots k_{\gamma_{\ell}}^{(\ell-1)}$$